

Theory Week 11

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Orga

Mechanics 3 CAB G59
WhatsApp group



- Find all doc's on my website (link in WhatsApp bio)
- remember to do the STACK
- Study Center
→ Wednesday 18:00-20:00 (ETFC 1)
- Summary videos from Dennis on Moodle

11	25/11/24 27/11/24	Lagrange equations, energy principles, stability vibrations, single-DOF free vibrations exercise 11: Lagrange equations, stability, eigenfrequencies	4-4.2 4.3-4.3.2
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4. Vibrations

≈ small oscillations around a stable equilibrium

all our eq. are for small oscillations

4.1 Lagrange equations

generalized velocity $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i^{\text{nc}}$ with $\mathcal{L} = T - V$

generalized force Q_i^{nc} with $Q_i^{\text{nc}} = \sum_{j=1}^N F_j^{\text{non-cons.}} \cdot \frac{\partial r_j}{\partial q_i}$

position of application point of non-conservative force

$Q_i^{\text{nc}} = 0$ if q_i is conservative

$q_1, q_2, q_3, \dots, q_n = \text{generalized DOFs}$

E.g. x for oscillations around 0 for rotations

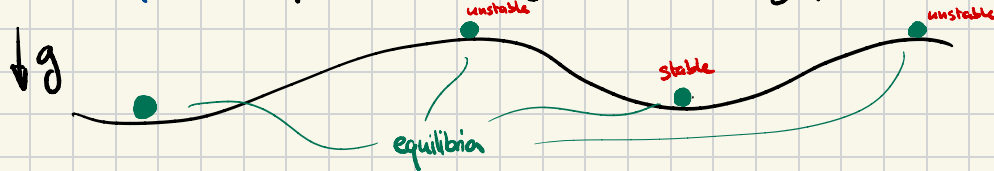
$F_{\text{cin}}, F_{\text{pot}}$

if there is only 1 DOF $\Rightarrow$ ignore all indices i

The Lagrange eq. provide a convenient alternative to LMB & MMB to obtain the eq. of motion.

4.2 Mechanical equilibrium

Def: The equilibrium of a conservative system is a stationary point of it's potential energy



equilibria of a conservative, static system with a single DOF q :

$$\frac{\partial V}{\partial q} = 0 \Rightarrow \begin{cases} \text{stable equilibrium} \Leftrightarrow \text{energy minimum} \Leftrightarrow \frac{\partial^2 V}{\partial q^2} > 0 \\ \text{unstable equilibrium} \Leftrightarrow \text{energy maximum} \Leftrightarrow \frac{\partial^2 V}{\partial q^2} < 0 \end{cases}$$

metastable equilibrium $\Leftrightarrow$ energy saddle point $\Leftrightarrow \frac{\partial^2 V}{\partial q^2} = 0$

$\rightarrow$ static $\Rightarrow T=0$
 $\rightarrow$ conservative $\Rightarrow Q^{\text{nc}} = 0$

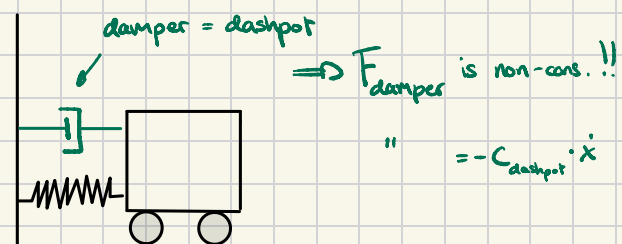
4.3 Single DOF Vibrations

prototypical eq. of motion for 1 DOF vibration

$$m\ddot{x} + c\dot{x} + kx = F$$

$\rightarrow$ linear ODE in time

- homogenous ODE if $F=0 \leftarrow$ free vibration
- inhomogenous ODE if $F \neq 0 \leftarrow$ forced vibration



"normalized form"

general form of the equation of motion for a single DOF:

$$\ddot{x} + 2\delta\dot{x} + \omega_0^2 x = f(t) \quad \text{with} \quad D = \frac{\delta}{\omega_0} = \text{Lehr's damping ratio}$$

equation of motion for a single-particle oscillator:

$$m\ddot{x} + c\dot{x} + kx = F(t) \quad \text{so} \quad \omega_0 = \sqrt{\frac{k}{m}}, \quad \delta = \frac{c}{2m}, \quad f(t) = \frac{F(t)}{m}$$

spring const. dashpot const.

☺ In dynamics we don't really have to solve these ODEs, we have derived solns that we use.

Solutions (for free vibrations)

ω_0 = **eigenfrequency** (natural / fundamental freq.)

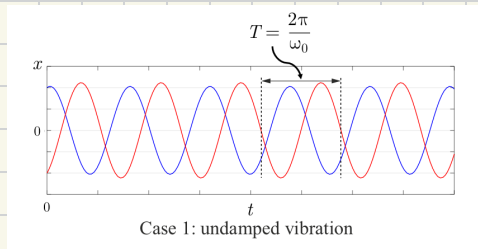
A = **amplitude**

$T = \frac{2\pi}{\omega_0}$ = **period**

①

general solution for free **undamped vibrations** ($D = 0$):

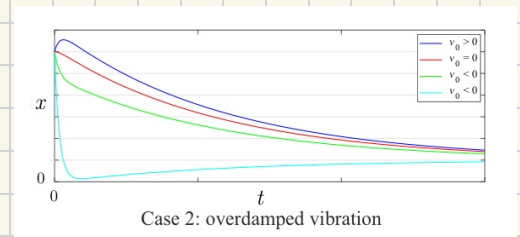
$$x(t) = A_1 \cos(\omega_0 t) + A_2 \sin(\omega_0 t) = A \sin(\omega_0 t + \varphi_0), \quad T = \frac{2\pi}{\omega_0}$$



②

general solution for free **overdamped vibrations** ($D > 1$):

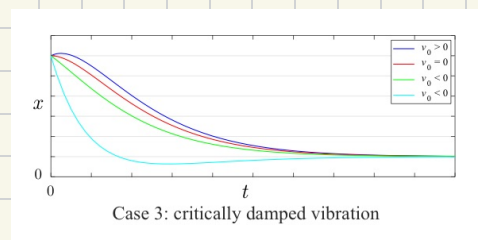
$$x(t) = A_1 e^{-(\delta + \sqrt{\delta^2 - \omega_0^2})t} + A_2 e^{-(\delta - \sqrt{\delta^2 - \omega_0^2})t} = A_1 e^{-\lambda_1 t} + A_2 e^{-\lambda_2 t}, \quad \text{where } \lambda_{1,2} = -\delta \pm \sqrt{\delta^2 - \omega_0^2}$$



③

general solution for free **critically damped vibrations** ($D = 1$):

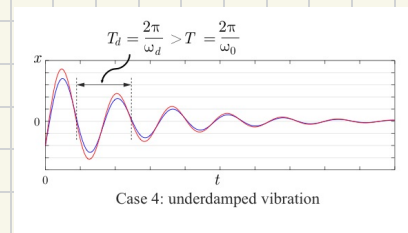
$$x(t) = A_1 e^{-\delta t} + A_2 t e^{-\delta t}$$



④

general solution for free **underdamped vibrations** ($0 < D < 1$):

$$x(t) = e^{-\delta t} [A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t)], \quad \omega_d = \sqrt{\omega_0^2 - \delta^2}, \quad T_d = \frac{2\pi}{\omega_d}$$



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